Zexin (Jason) Xu

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Research Interests

My research interests lie in multimodal learning and AI agents. I currently focus on **egocentric multimodal agents** that understand what a user is doing from first-person video and audio and offer help **proactively**. More broadly, I am interested in **post-training** and **alignment** for these systems, with applications to **educational technology**.

Education

The University of Texas at Dallas — Aug 2024 – Present
Ph.D. Candidate in Computer Science — *Advisor: Dr. Yapeng Tian*

The Ohio State University — Aug 2021 – May 2023
M.S. in Computer Science and Engineering — *Advisor: Dr. Yu Su*

The Ohio State University — Jan 2018 – Dec 2020
B.S. in Computer Science and Engineering / *Magna Cum Laude, Dean's List*

Experience

Project Lead, Closed-Loop Benchmarking of Proactive AI Assistants — Jul 2026 – Present
The University of Texas at Dallas, advised by Dr. Yapeng Tian — *Richardson, TX*

- Leading a benchmark that judges proactive vision-language assistants by **whether their warnings actually prevent mistakes**, instead of by matching annotated timestamps on replayed video, where the outcome never depends on what the assistant says.
- Built **closed-loop simulation** environments in two domains, driving in **CARLA** (Bench2Drive scenarios) and household tasks in **BEHAVIOR / OmniGibson** on NVIDIA Isaac Sim, where an agent hears and acts on the assistant's warnings in real time.
- Designed a **two-arm counterfactual evaluation protocol**: each scenario is rolled out deterministically with and without the assistant, and the last moment a warning can still change the outcome is computed by counterfactual rollout rather than hand annotation.
- Defined outcome-based **benchmarking metrics**, including prevention rate, warning precision, graded lead-time credit, and risk ratio, for evaluating streaming VLM assistants.

Founder — Mar 2026 – Present
Juku Academy (juku-academy.vercel.app)

- Founding an early-stage academy (in development) that teaches students aged 12–18 **design thinking with coding agents**: framing real problems, prototyping, and iterating on their own projects, with an **LLM mentor** guiding each student from problem definition to a working prototype and a shareable portfolio.
- Building the platform end to end with **Next.js, TypeScript, Supabase**, including an LLM orchestration layer that tracks student reasoning and progress, and an AI-generated parent report with verbatim citations into student chats and milestones.

Research Intern — Sep 2025 – Mar 2026
Intelligent Robotics and Vision Lab (IRVL), The University of Texas at Dallas — *Richardson, TX*

- Conducted research in **offline reinforcement learning** for robotic manipulation under the supervision of Dr. Yu Xiang, learning policies from pre-collected datasets.
- Developed and evaluated manipulation policies in simulation using **Gazebo** and **NVIDIA Isaac Gym**.

Team Lead — Aug 2024 – Jun 2025
The University of Texas at Dallas (Amazon Nova AI Challenge: Trusted AI) — *Richardson, TX*

- Led team ASTRO representing UTD, participated in the first Nova AI Trusted AI challenge on Code LLM red teaming and defense, selected as one of **5 red teams globally**.
- Achieved **2nd place in 2 tournaments** and finalist status; published paper in Amazon Science proceedings.
- Developed red teaming pipeline to trigger defense models on generating malicious/vulnerable code using **CodeGuru, Bedrock, Cobot** and other AWS services.

Software Engineering Lead Nov 2023 – Mar 2026
University of California, Irvine (Part Time) *Irvine, CA*

- Co-led TimeWise educational conversational agent project with researchers from **UCI, NTU, and JHU**.
- Selected as **finalist in 2024 Tools Competition** among ∼2,000 submissions; built React-based Chrome extension with full AWS infrastructure.

Research Assistant, Software Engineering Lead Aug 2022 – Jun 2023
The Ohio State University (Amazon Alexa Prize SimBot Challenge) *Columbus, OH*

- Led engineering team for SalsaBot in the first Alexa Prize SimBot Challenge on **embodied agents**, architecting state-driven dialogue system with language macro module using AWS.
- Elevated team rating from **2.36 to 3.64 out of 5.00**, achieving top-three standing in final month through comprehensive testing methodologies.

Data Scientist Intern May 2022 – Aug 2022
AI Camp *Palo Alto, CA*

- Led a team of **15+ students** building NLP web applications with **GPT-2 and Flask** (average rating **4.53/5.00**), and contributed to the NLP, deep learning, and Flask curricula.

Publications

Note: * denotes equal contribution. Full list: Google Scholar

1. **AnchorCard: Attributed Card Generation from Web Documents**
   Zexin Xu, Haoxuan Zhang, Manraj Singh, Gopika Murali, Gagandeep Singh, Lizal Adhikari, Mehri Sattari, Hamid Rouhani, Anthony C. Garza, Vishwaa Shah, Ivin Issac Georgi, Akib Jawad Ononto, S M Tahmid Siddiqui, Akshay Vyas, Jiasheng Xu, Daud Aftab Masih, William T. Doan, Ziqi Liu, Yapeng Tian, Haihua Chen
   Under review, 2026
2. **Do Joint Audio-Video Generation Models Understand Physics?**
   Zijun Cui, Xiulong Liu, Hao Fang, Mingwei Xu, Jiageng Liu, Zexin Xu, Weiguo Pian, Shijian Deng, Feiyu Du, Chenming Ge, Yapeng Tian
   Conference on Neural Information Processing Systems (NeurIPS), 2026 (Poster)
3. **COMET: Closed-Loop Orchestration for Malicious Elicitation Techniques in Code Models**
   Zexin Xu, Tingxi Li, Ravishka Shemal Rathnasuriya, Zihe Song, Jun Ren, Bhavesh Mandalapu, Soroush Setayeshpour, Xinya Du, Wei Yang
   Amazon Nova AI Challenge Proceedings, 2025
4. **LLM4SR: A Survey on Large Language Models for Scientific Research**
   Ziming Luo*, Zonglin Yang*, Zexin Xu, Wei Yang, Xinya Du
   ACM Computing Surveys (CSUR), to appear, 2026
5. **SoK: Efficiency Robustness of Dynamic Deep Learning Systems**
   Ravishka Rathnasuriya, Zexin Xu*, Tingxi Li*, Zihe Song*, Mirazul Haque, Simin Chen, Wei Yang
   USENIX Security, 2025
6. **Beyond Pass or Fail: A Multi-Dimensional Benchmark for Mobile UI Navigation**
   Dezhi Ran, Mengzhou Wu, Hao Yu, Yuetong Li, Jun Ren, Yuan Cao, Xia Zeng, Haochuan Lu, Zexin Xu, Mengqian Xu, Ting Su, Liangchao Yao, Ting Xiong, Wei Yang, Yuetang Deng, Assaf Marron, David Harel, Tao Xie
   arXiv preprint, 2025

7. **Leveraging Generative AI in Designing and Delivering Individualized Responsive Feedback for Pre-service Teachers in Higher Education**
   Jingwen He, Tingting Li, Zexin Xu, Kui Xie
   Artificial Intelligence and Human Agency in Education: Volume Two: AI for Equity, Well-Being, and Innovation in Teaching and Learning, 2025
8. **SalsaBot: Towards a Robust and Generalizable Embodied Agent**
   Chan Hee Song, Jiaman Wu, Ju-Seung Byun, Zexin Xu, Vardaan Pahuja, Goonmeet Bajaj, Samuel Stevens, Ziru Chen, Yu Su
   Embodied AI Workshop at CVPR, 2023
9. **Towards a Robust and Generalizable Embodied Agent**
   Chan Hee Song, Jiaman Wu, Ju-Seung Byun, Zexin Xu, Vardaan Pahuja, Goonmeet Bajaj, Samuel Stevens, Ziru Chen, Yu Su
   Amazon Science, 2023
10. **Exploring the Role of Artificial Intelligence in Facilitating Assessment of Writing Performance in Second Language Learning**
    Zilu Jiang, Zexin Xu, Zilong Pan, Jingwen He, Kui Xie
    Languages, 2023
11. **Measuring Elementary Students' Behavioral Engagement in Web-based Science Inquiry Learning**
    Jingwen He, Bihui Jin, Zexin Xu, Danhui Zhang
    Journal of Online Learning Research, 2022

Teaching

Guest Lectures

"A Tale of Two Vibe Coders" *(University of North Texas)*	*Jul 2026*
"From Base to Best: The LLM Post-Training Series" *(UT Dallas, 3 sessions)*	*Oct–Nov 2025*
"Reliable LLM via Guardrails" *(UT Dallas)*	*Feb 2025*

The University of Texas at Dallas — *Graduate Teaching Assistant*

CE4337 Programming Language Paradigms	*AU 24, SP 25, AU 25, AU 26*
CS6375 Machine Learning	*SP 26*
CS4375 Introduction to Machine Learning	*AU 25*
CS5343 Algorithm Analysis and Data Structures	*SU 25*
SE3377 Systems Programming	*SU 25*

The Ohio State University

CSE6521 Advanced Artificial Intelligence *(Graduate Teaching Associate)*	*AU 22*
CSE3521 Artificial Intelligence I: Basic Techniques *(Undergraduate TA)*	*AU 19*
CSE2321 Foundations I: Discrete Structures *(Undergraduate TA)*	*SP 20*

Awards & Grants

Amazon Nova AI Challenge (Trusted AI) Grant — Jan 2025 – Jul 2025
Team award to UTD Team ASTRO (Team Lead) — $250,000

Amazon Nova AI Challenge (Trusted AI) AWS Research Credit — Oct 2024 – Jul 2025
Team award to UTD Team ASTRO (Team Lead) — $150,000 per month, $1,350,000 total

Skills

Languages: Python, Java, C/C++, C#, SQL, JavaScript, R
Simulation: CARLA, NVIDIA Isaac Sim, OmniGibson, NVIDIA Isaac Gym, Gazebo, MuJoCo
Libraries: PyTorch, HuggingFace Transformers, LangChain, OpenCV, Scikit-learn
Frameworks & Tools: AWS, Next.js, Node.js, Flask/Django, Docker, Git, CUDA
Databases: MongoDB, MySQL, PostgreSQL